Nuvoton NuML_Studio Quick Start with Edge Impulse Project

Description

This document explains how to build a TinyML project with Edge Impulse and NuML_Studio, and deploy it on the NuMaker-M55M1 using image classification as an example.

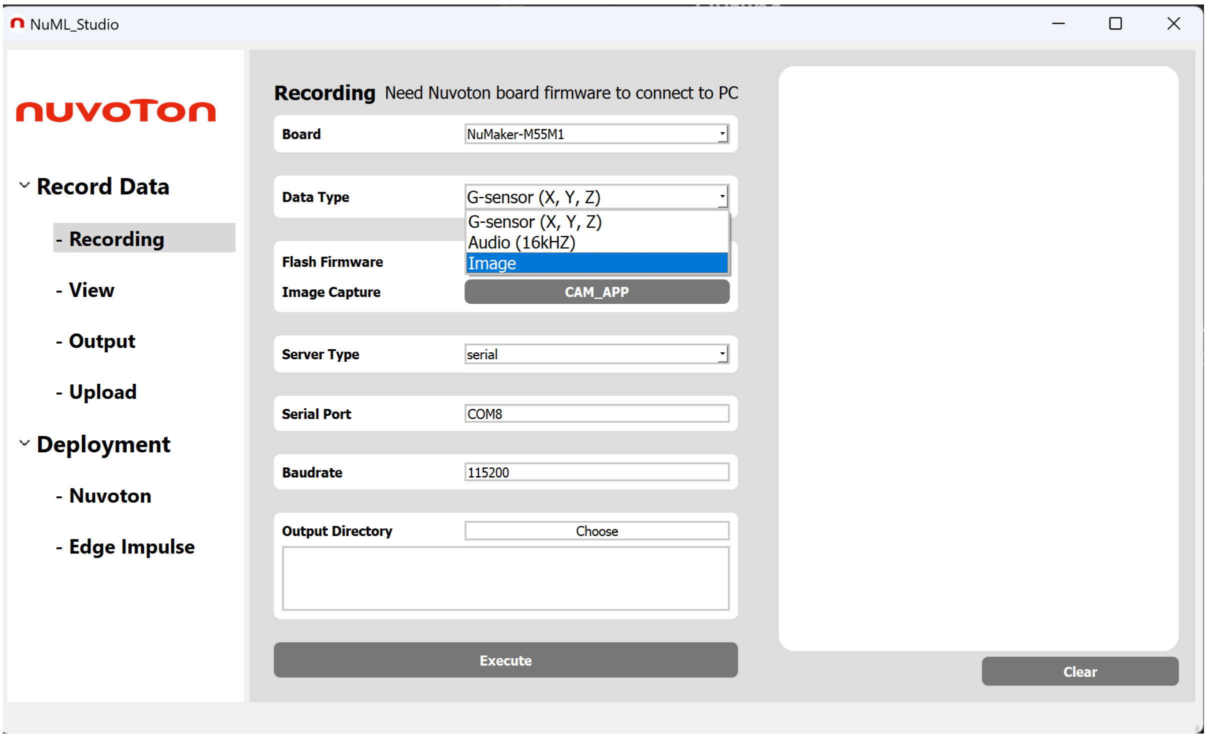
Outline

Getting Started

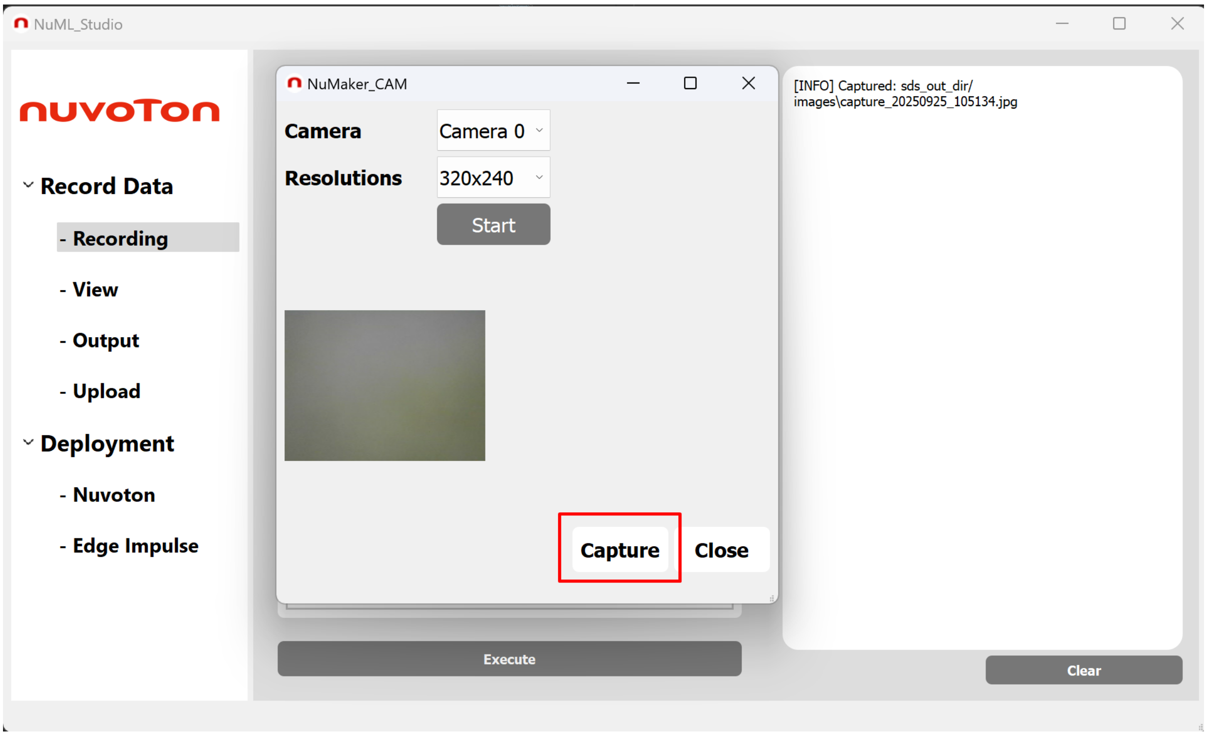
Collecting Data by NuML_Studio

1. Flash the data collection firmware

   - Select `Image`, click `Flash Firmware` and wait for completion.

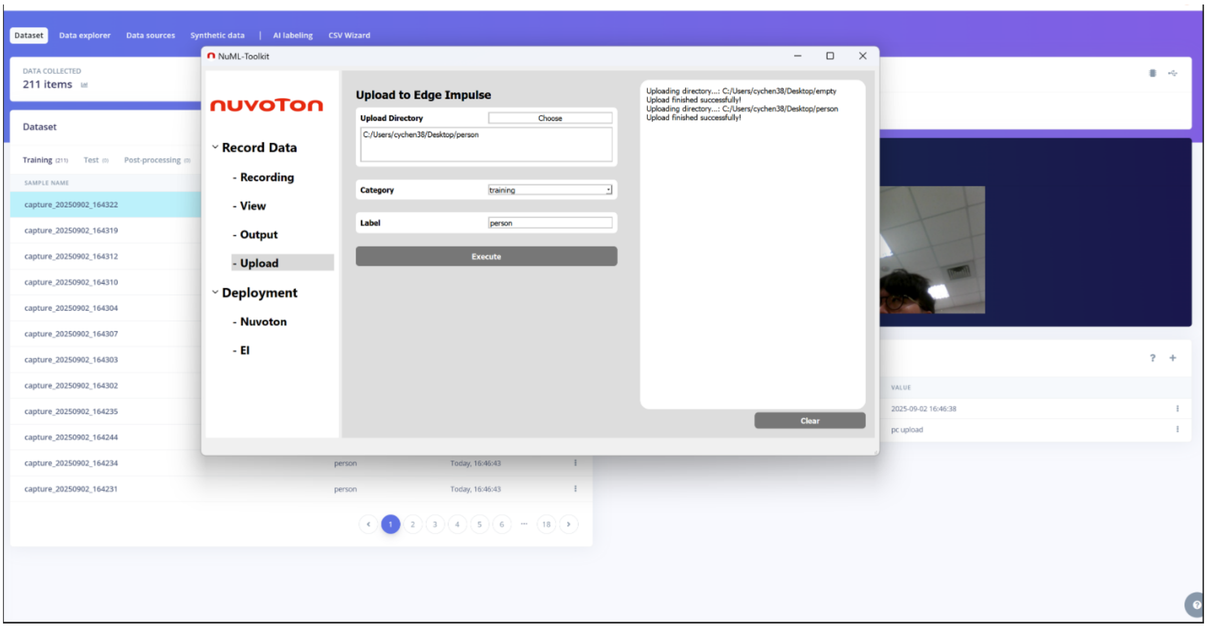


   - Click `Image Capture` to take and save images.



2. Upload data to Edge Impulse

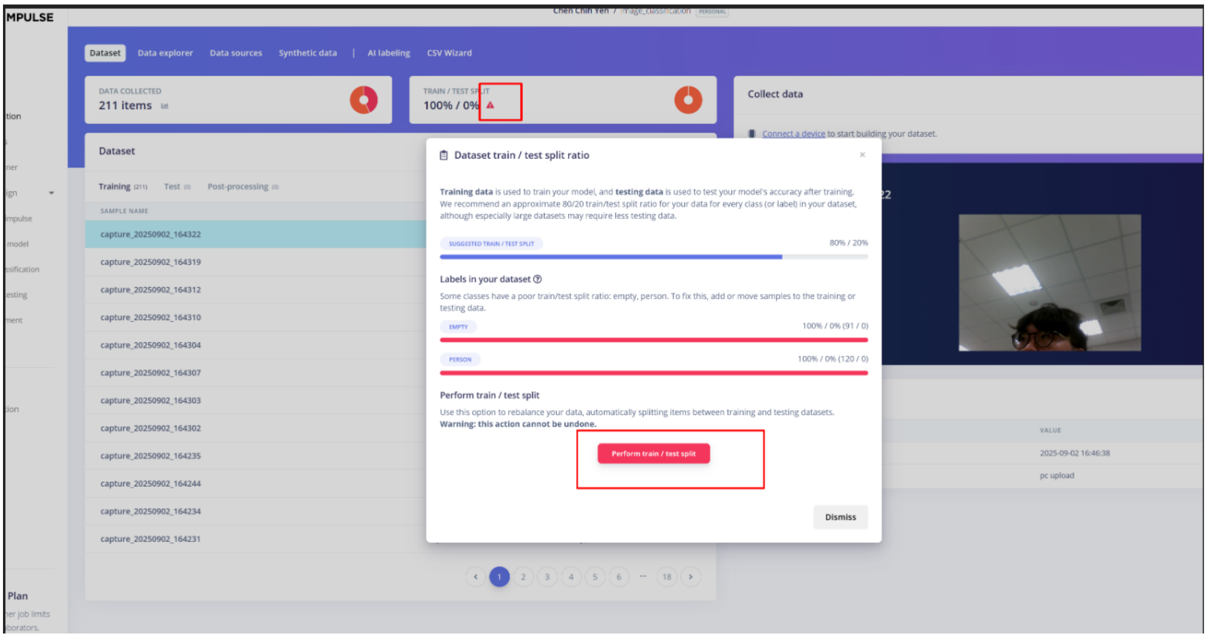
   - Copy your project API-Key from Edge Impulse `Dashboard`, `Keys` and paste it into `NuML_Studio_vX.X.X\API_key.txt`.
   - Use NuML_Studio to upload your collected data folder, or upload directly via Edge Impulse `Data acquisition` page.



Training by Edge Impulse

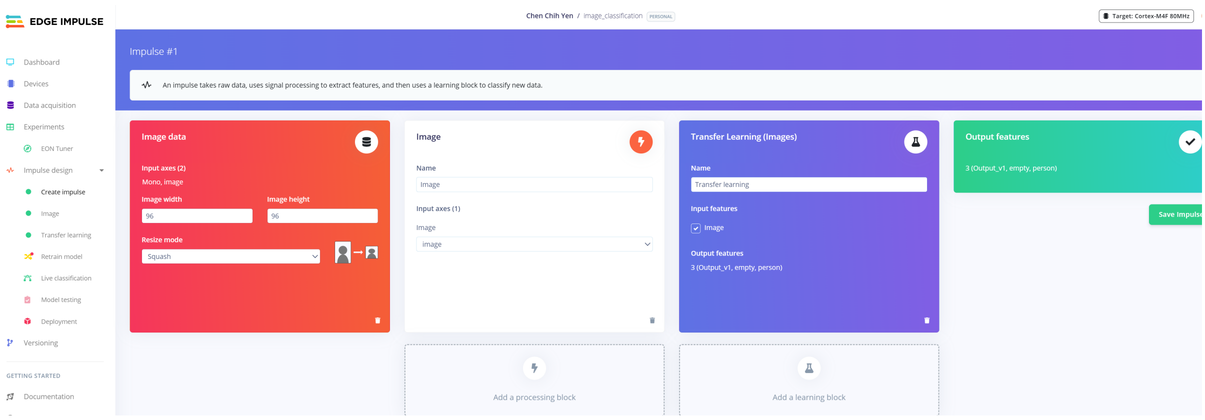
1. Data Acquisition

   - Go to the `Data acquisition` tab in Edge Impulse.
   - Review and update labels if needed.
   - Split the collected data into training and testing datasets.



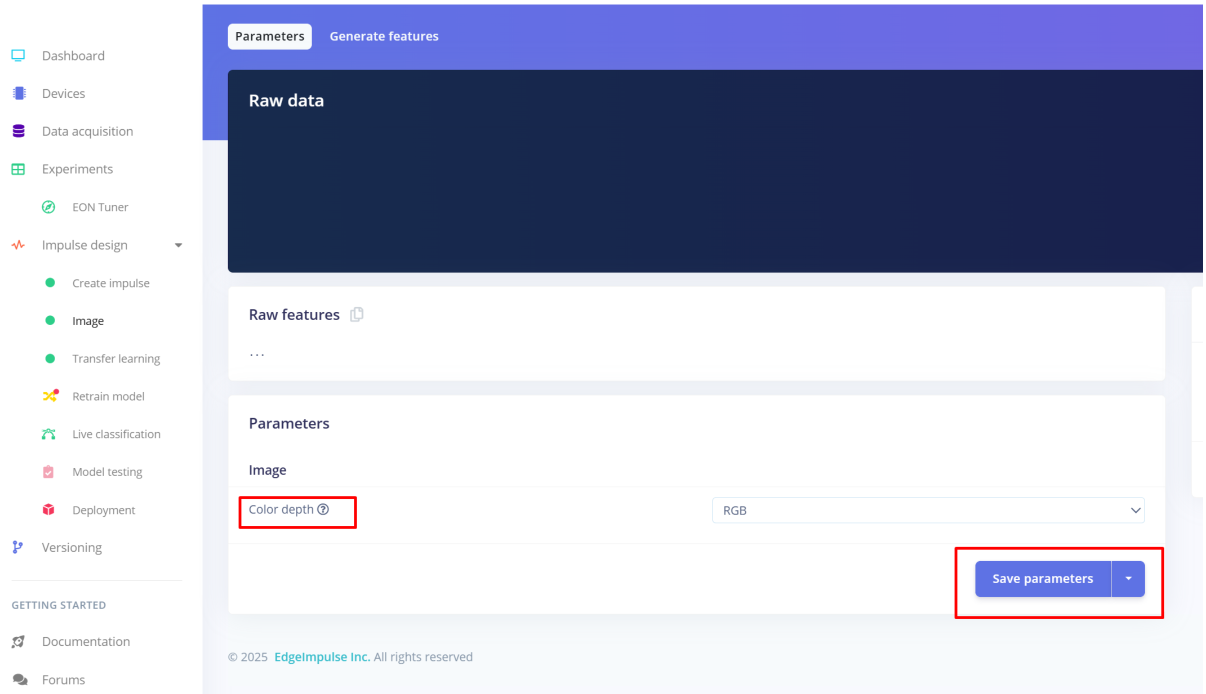
2. Create Impulse

   - Go to the `Create Impulse` tab.
   - Configure your ML pipeline (preprocessing, learning block, classifier).
   - For Image input, set the `Resize mode` to **Squash**

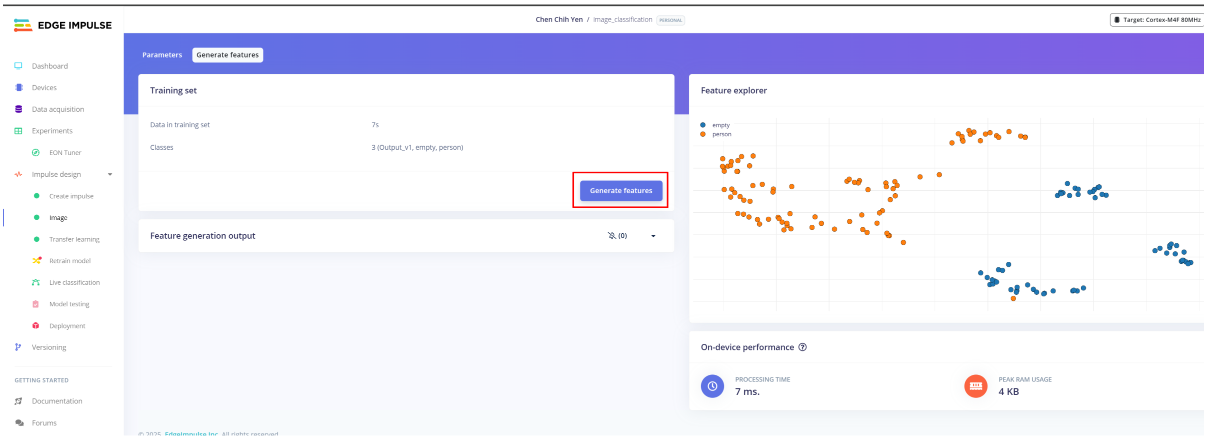


3. Set Parameters and generate Features

   - Go to the `Image` tab.
   - Set Parameters and click `Save parameters`.

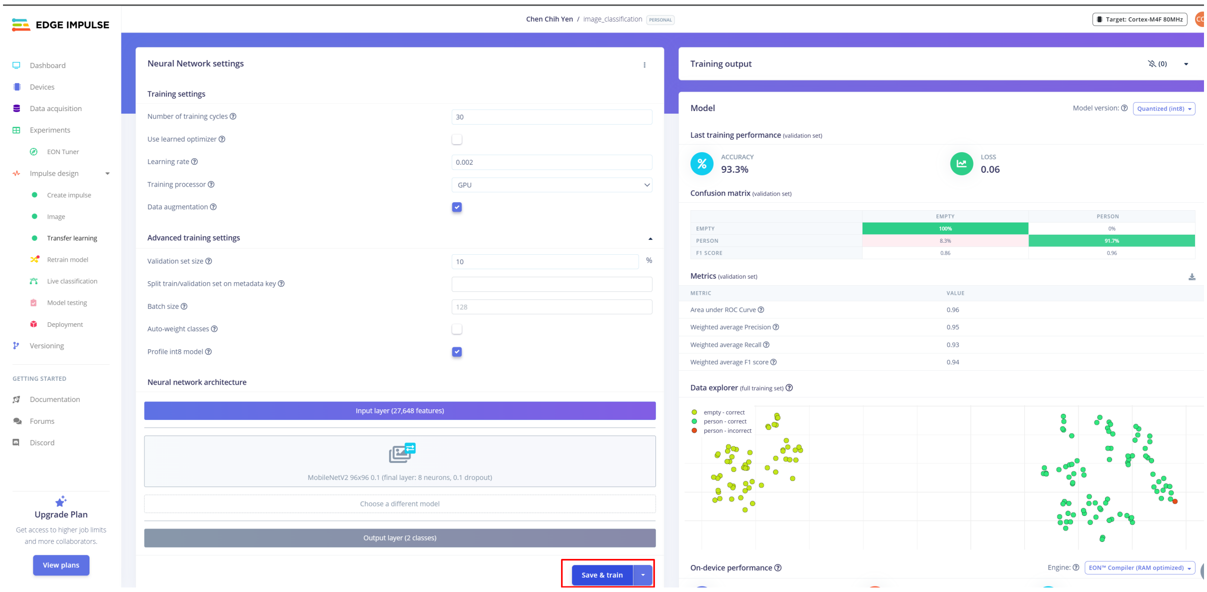


- Then click `Generate features` to extract feature vectors from your dataset.



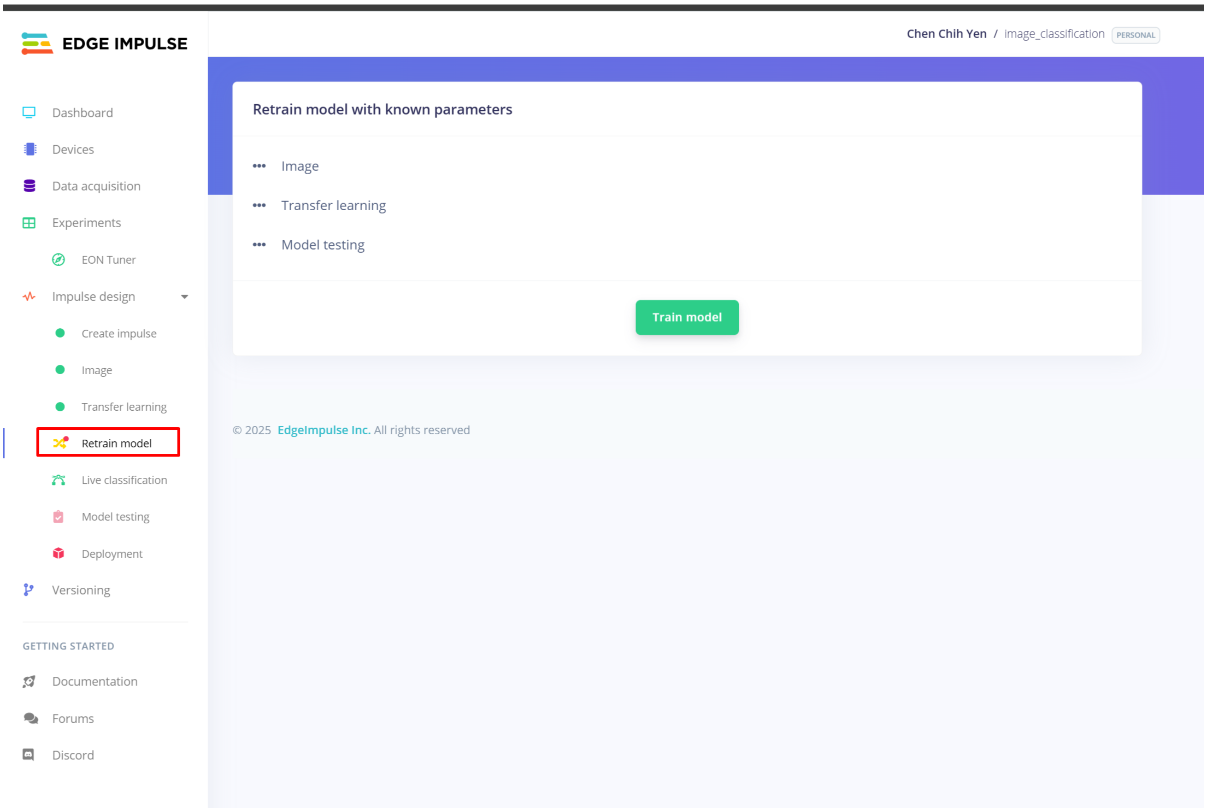
4. Start Training

- Go to the `Transfer Learning` or `Learning` tab.
- Select the model type.
- Adjust training parameters (epochs, learning rate, validation split, etc.).
- Click `Start training` to begin model training.



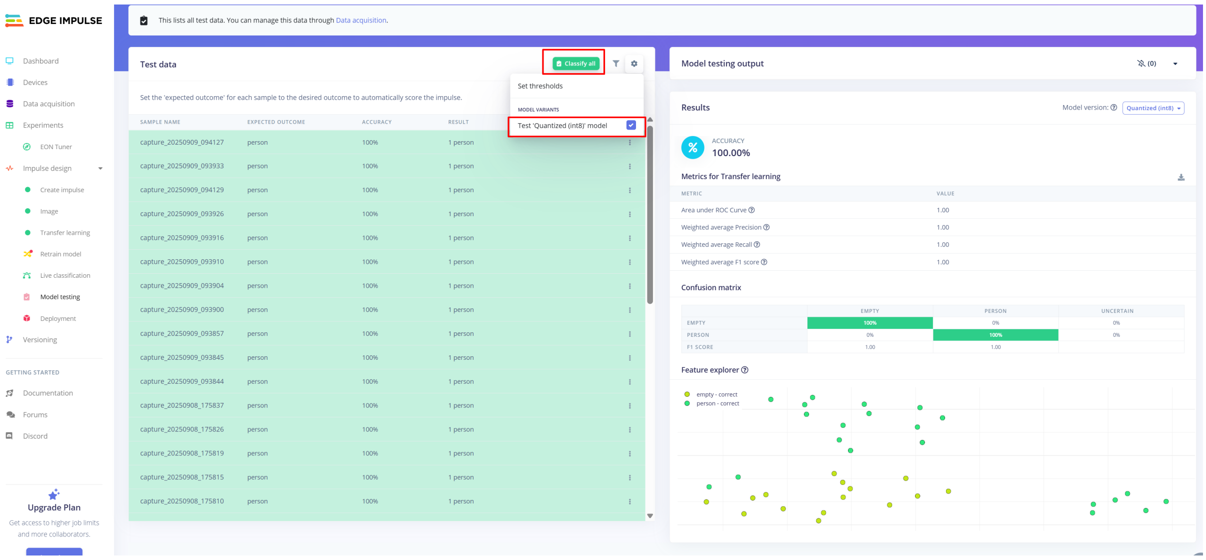
5. Retrain model

   - Go to the `Retrain model` tab.



6. Test the model

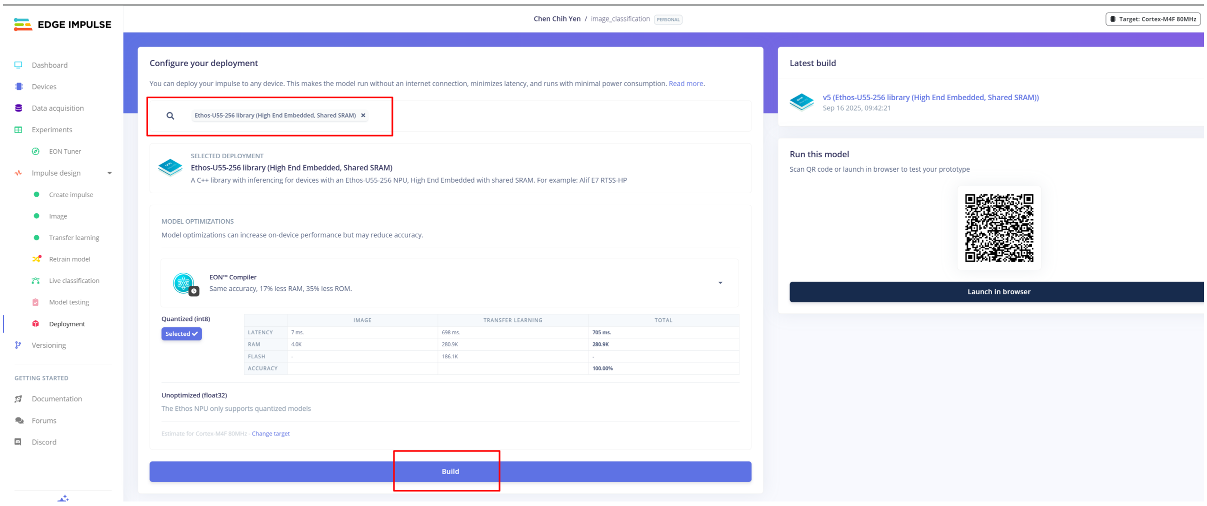
   - Go to the `Model testing` tab.
   - Please choose **Test Quantized model**. This will test the quantized model using all the test data.



Deployment

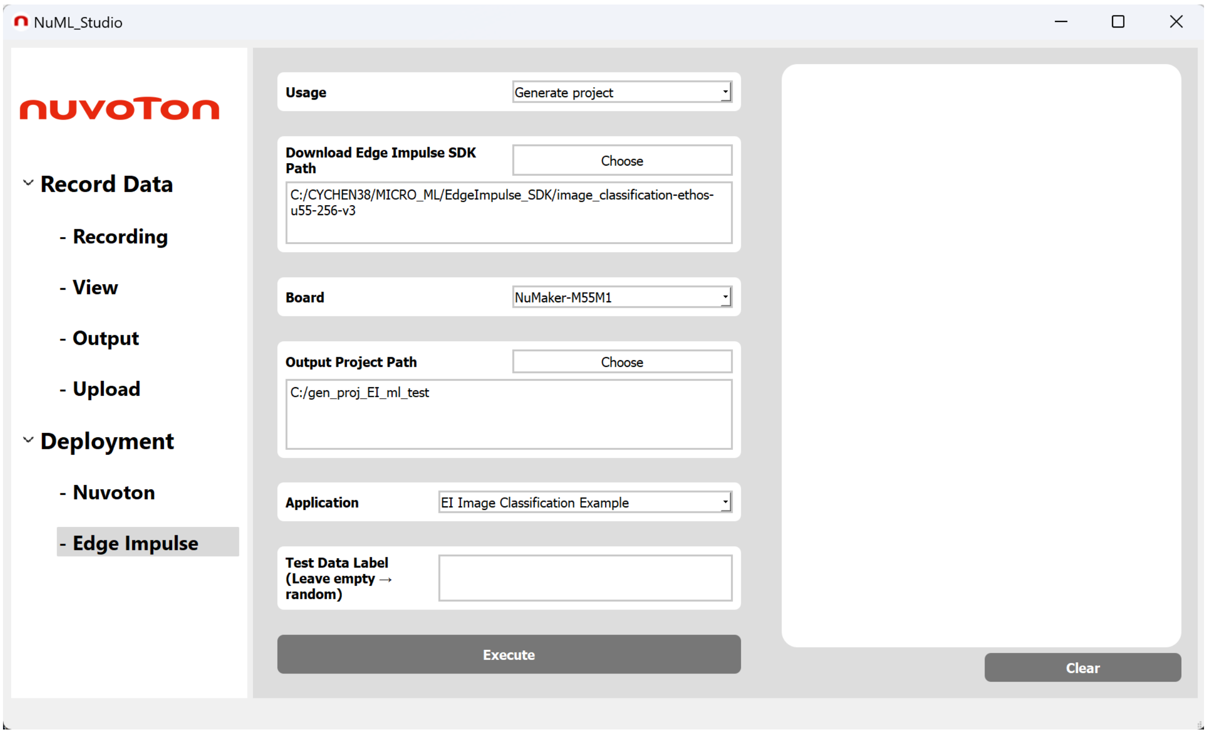
1. Build the deployment model and SDK

   - Go to the `Deployment` tab on the Edge Impulse project page.
   - Select **ethosu-U55 256 library** and build it.



2. Run inference firmware on NuMaker-M55M1

   - After downloading and unzipping the SDK, use NuML-Studio to create the code project.
   - In NuML-Studio, select `- Deployment`, `- Edge Impulse`.



- The final step is to open the Keil or VSCode CMSIS project generated by NuML-Studio, then build and flash the NuMaker-M55M1 ML firmware with your Edge Impulse model.